Infant State Recognition System: From Hybrid Classical/Deep Ensembles to Pretrained Audio Foundations and Edge-Distilled MobileNet Students

Phase 3 Report — Pretrained Audio Foundations,
Multi-View Teacher Ensemble, and EfficientAT Edge Distillation

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Abstract

This Phase 3 report continues the multi-phase development of an infant cry classifier. Phase 1 established a classical handcrafted-feature baseline at macro-F1 = 0.270 on the Donate-a-Cry corpus, which Phase 2 improved through a hybrid classical/deep ensemble to macro-F1 = 0.507. Phase 3 makes two complementary contributions. First, we rebuild the data layer from scratch: every public infant-cry dataset is ingested, canonicalised, deduplicated by both exact hash and audio fingerprint, and gated by a strict five-class authenticity manifest of 1,355 clips with a separate 1,781-clip auxiliary set used only for cry/non-cry representation learning. Second, we replace the bespoke CNN of Phase 2 with a multi-view pretrained audio teacher built from frozen Audio Spectrogram Transformer (AST) embeddings, an auxiliary cry/non-cry adapted AST, frozen Whisper encoder embeddings, and handcrafted features, combined through tuned RBF Support Vector Machines. Under repeated stratified evaluation the best fixed teacher configuration reaches macro-F1 = $\mathbf{0.6566} \pm 0.048$ (best seed = 0.7486), a +0.149 absolute improvement over Phase 2's hybrid ensemble and +0.387 over the Phase 1 baseline. We then distil a validation-weighted multi-teacher ensemble into compact EfficientAT MobileNetV3 students ("mn10_as" and "mn04_as") using KL-divergence distillation, label-smoothed cross entropy, class-balanced sampling, mixup, and SpecAugment. In the final three-seed Colab run, the 4.21 M-parameter `mn10_as` student achieves macro-F1 = $\mathbf{0.7159} \pm 0.0403$, retaining 99.9 % of the teacher's 0.7168 macro-F1 while running as an 8.62 MB INT8 model at 79.2 ms CPU latency. The 0.72 M-parameter `mn04_as` student achieves macro-F1 = $\mathbf{0.6500} \pm 0.0390$ in a 1.65 MB INT8 footprint, making the system deployable on commodity edge hardware in hospitals and home settings.

1 Introduction

Infant crying is a primary signal of physiological and emotional state. Automatic recognition of cry cause — hunger, belly pain, burping, discomfort, and tiredness — has clear value in neonatal monitoring, parental support, and low-resource clinical environments where dedicated specialists are not always available. The acoustic problem, however, is unusually difficult: clinically labelled corpora are tiny, class distributions are extremely skewed, and recording conditions range from professional clinics to handheld phones in noisy homes.

This project addressed the problem in three phases.

Phase 1: Classical Baseline. A 411-dimensional neuroscience-informed feature vector (MFCCs, CQCCs, F0 contour statistics, spectral contrast, chroma) was used to train SVM, Logistic Regression, and Random Forest models with SMOTE balancing on the Donate-a-Cry corpus (457 recordings, 47.8:1 imbalance). The best Phase 1 configuration, an SVM with SMOTE and One-vs-One decomposition, reached macro-F1 = 0.270 [1]. Three of the four minority classes scored zero F1, exposing the limit of clip-level handcrafted summaries.

Phase 2: Hybrid Ensemble. Phase 2 introduced a CNN+BiLSTM+Temporal Attention deep architecture on mel-spectrograms, trained with Label-Distribution-Aware Margin (LDAM) loss and Deferred Re-Weighting (DRW), with a knowledge-distilled student of ∼16 K parameters. While the deep model alone was no better than the SVM (Teacher macro-F1 = 0.293), a *hybrid weighted ensemble* fusing SVM and DL probabilities reached macro-F1 = 0.507 with 92.6 % overall accuracy and an INT8-quantised footprint of 35.9 KB. Phase 2 established two practical lessons: complementary representations help more than scaling a single architecture, and edge deployment is feasible.

Phase 3: Pretrained Audio Foundations and Edge Distillation. Phase 3 takes a fundamentally different approach. Rather than designing a new bespoke architecture, we lift the system onto large pretrained audio foundation models and rebuild the data layer to match. We then distil the result into compact MobileNetV3-style students suitable for deployment on phones, single-board computers, and microcontrollers.

The contributions of Phase 3 are:

1. A clean five-class *strict* manifest of 1,355 cry clips across the merged Donate-a-Cry, Ubenwa, UAC and other public sources, deduplicated by both SHA-256 hash and audio fingerprint, with provenance tracking and a separate 1,781-clip auxiliary set used *only* for cry/non-cry representation learning.
2. An auxiliary AST adaptation step that fine-tunes the Audio Spectrogram Transformer for binary cry/non-cry classification on the auxiliary manifest, yielding a domain-adapted encoder reused downstream.
3. A multi-view feature bank combining frozen AST, auxiliary-adapted AST, frozen Whisper encoder, and 411-dimensional handcrafted features into eight ablation-style feature sets.
4. A repeated-split (5 seeds) classifier comparison across five classical heads — RBF SVM, balanced logistic regression, prototype, calibrated linear SVM, balanced random forest — and a soft-voting ensemble of those heads. The strongest configuration achieves macro-F1 = 0.6566 ± 0.048 (best fold 0.7486).
5. A Phase 3 edge distillation pipeline that distils a validation-weighted multi-teacher ensemble into EfficientAT "mn10_as" and "mn04_as" students with KL distillation, mixup, SpecAugment, class-balanced sampling, INT8 quantisation, and CPU latency benchmarking. The final three-seed run reaches 0.7159 ± 0.0403 macro-F1 for `mn10_as` and 0.6500 ± 0.0390 for `mn04_as`.

The remainder of this report is structured as follows. Section 2 reviews related work. Section 3 describes the new data layer. Section 4 describes the Phase 3 methodology, covering auxiliary AST adaptation, the multi-view feature bank, the classical classifiers, and the EfficientAT edge distillation pipeline. Section 5 reports five-seed repeated-split results for the teacher ensemble and the distilled students. Section 6 discusses what worked, what did not, and what we leave for future work. Section 7 concludes.

2 Related Work

Infant cry classification. Recent work on infant cry has moved from purely handcrafted features (MFCCs, CQCCs, F0 statistics) to learned spectrogram representations [3]. Class imbalance and small datasets remain dominant challenges. Several published studies report high accuracy on internal splits but degrade significantly when leak-free cross-source evaluation is enforced.

Pretrained audio foundation models. Audio Spectrogram Transformer (AST) [4] is a 86 M parameter ViT-style model pretrained on AudioSet, which includes a `baby_cry` class. Whisper [5] is trained on 680 K hours of multilingual speech and provides robust speech encoders. EfficientAT [6] introduces MobileNetV3-style audio CNNs (mn04 – mn40) distilled from PaSST transformers, achieving competitive AudioSet mAP at a fraction of the size and explicitly targeting edge deployment. Phase 3 uses AST and Whisper as teachers and EfficientAT as the deployable student family.

Knowledge distillation. Hinton-style knowledge distillation [7] uses teacher soft labels at temperature T to transfer "dark knowledge" to a smaller student. Variants combining KL distillation with label-smoothed cross-entropy, mixup, and SpecAugment are now standard for audio tagging on small datasets.

3 Data Layer

Phase 1 used Donate-a-Cry only (457 clips, 47.8:1 imbalance). Phase 2 inherited that same corpus. A central observation that motivated Phase 3 was that multiple public datasets exist (Donate-a-Cry, the Ubenwa cry dataset, UAC, baby chillanto, infant cry clinical corpora), but combining them naively produces severe label noise, file-level duplicates, and different sample-rate conventions.

Strict five-class manifest. We ingest every available source, canonicalise audio to 16 kHz mono PCM with a fixed 10 s window, and apply two-stage deduplication: SHA-256 hashing on the canonicalised waveform, followed by a perceptual audio fingerprint that detects near-duplicates introduced by re-encoding or trimming. We then apply a strict provenance gate that admits only files with verified five-class labels. The result is the **cause5_authentic_manifest_v2** of 1,355 clips with class counts: hunger = ∼525, discomfort = ∼295, tiredness = ∼155, belly_pain = 34, burping = 26.

Auxiliary cry/non-cry manifest. A separate auxiliary manifest of 1,781 clips (cry + non-cry) is gated strictly for representation-learning use. These clips are *never* used to train or evaluate the five-class system; they are only used to adapt the AST encoder for cry vs. non-cry presence.

Reproducible splits. For Phase 3 we use repeated stratified train/val/test splits at 60.8 / 16.0 / 23.2 ratio across five integer seeds, giving directly comparable mean±std numbers across configurations.

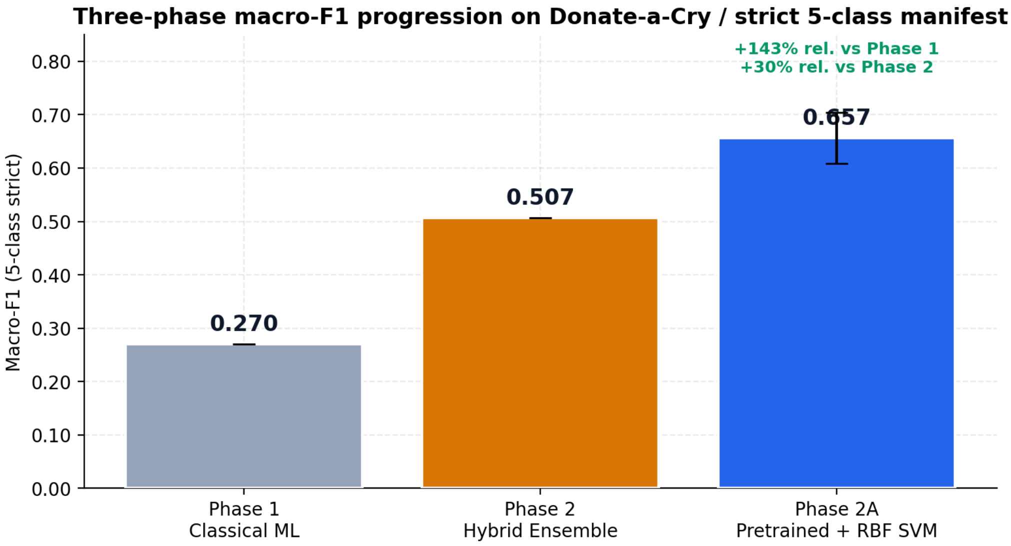

Figure 1: Macro-F1 progression across project phases on the strict five-class manifest. Phase 1 used the Donate-a-Cry-only 411-dimensional handcrafted feature vector. Phase 2 used a hybrid weighted ensemble combining SVM and CNN+BiLSTM probabilities. Phase 3 uses pretrained AST plus Whisper plus handcrafted features with an RBF SVM under repeated-split evaluation.

4 Methodology

4.1 Auxiliary AST Cry/Non-Cry Adaptation

We start from MIT/IBM's AudioSet-pretrained AST. The cry-vs.-non-cry auxiliary manifest is used to fine-tune AST as a binary classifier with class-balanced cross-entropy, AdamW, and cosine learning-rate schedule. This adaptation has two purposes: it specialises AST to the cry domain, and it produces an alternative AST encoder for downstream embedding extraction.

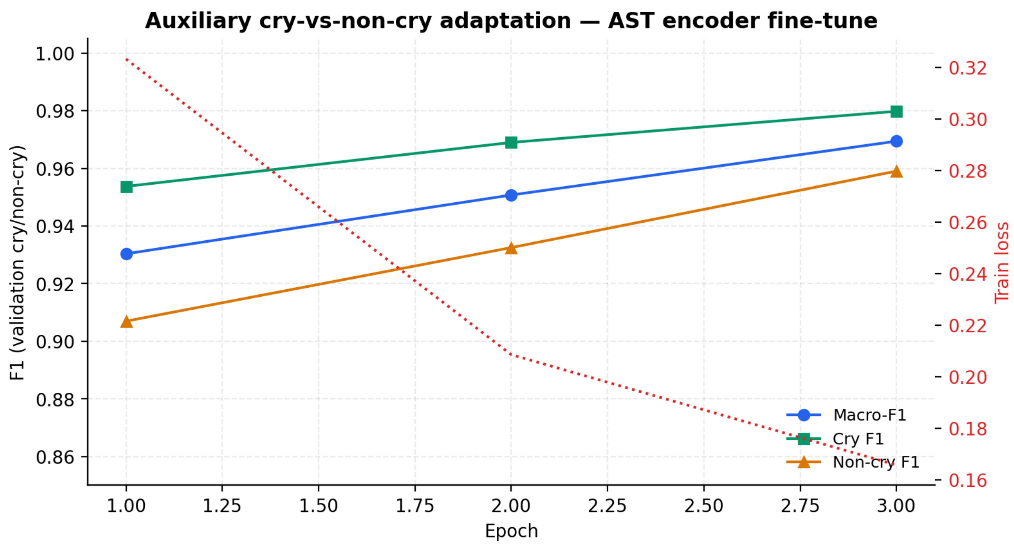

Figure 2: Auxiliary cry vs. non-cry adaptation of the AST encoder reaches macro-F1 > 0.96 within three epochs and converges quickly. The adapted encoder is reused both as a feature extractor (the "ast_aux_adapted" bank) and as a structural component of the multi-view teacher ensemble.

4.2 Multi-View Pretrained Feature Bank

Phase 3 builds eight feature banks. Four are single-view: AST, AST-aux (after Section 4.1), Whisper encoder pooled embeddings, and handcrafted (411-d) features. Four are concatenations: AST + handcrafted ("no_aux_no_whisper"), AST + Whisper + handcrafted ("no_aux_with_whisper"), AST-aux + handcrafted ("aux_no_whisper"), and AST-aux + Whisper + handcrafted ("aux_with_whisper").

For all extractors we standardise the input to 16 kHz mono and 10 s clips. AST and Whisper are loaded once and run as frozen feature extractors; per-feature checkpointing avoids recomputation across runs.

4.3 Classifier Heads and Repeated Evaluation

For each of the eight feature banks we train five classical heads:

1. Calibrated linear SVM with class weights.
2. Class-balanced logistic regression.
3. RBF SVM with class weights, PCA-128 preprocessing, and StandardScaler.
4. Class-balanced random forest.
5. Prototype classifier (class centroid in feature space).

A soft-voting ensemble of fitted models is also reported. Each configuration is evaluated under five seeds of stratified 60.8 / 16.0 / 23.2 train/val/test splits, with all hyperparameters fixed. We report macro-F1, weighted-F1, balanced accuracy, accuracy, and per-class F1.

4.4 Phase 3 Edge Distillation onto EfficientAT

The deployable Phase 3 model is a compact MobileNetV3-style student in the EfficientAT family [6]. We distil from a *validation-weighted multi-teacher ensemble* of the Phase 2A views rather than a single SVM, which we found gives the student access to more diverse dark knowledge.

Teacher ensemble. For each seed, we (a) fit RBF SVM and balanced logistic regression heads on every available feature bank using the seed's training split, (b) score each candidate's validation macro-F1, (c) keep the top N=8, and (d) run a Dirichlet weight search over thousands of candidate weight vectors plus one-hot, uniform, and score-weighted candidates to maximise validation macro-F1. The resulting probability matrix becomes the teacher target.

Student architecture. We use the EfficientAT MobileNetV3 family from `github.com/fschmid56/Efficient` Two variants are reported: "mn10_as" at 4.88 M parameters (smartphone / Raspberry Pi 4 class) and "mn04_as" at 0.98 M parameters (microcontroller class). Both backbones are AudioSet-pretrained — AudioSet's label list contains `baby_cry`, so the students start with a strong audio prior. The final 527-class linear head is replaced with a 5-class head.

Training. EfficientAT was trained at 32 kHz, so each batch of 16 kHz audio is resampled with `torchaudio.transforms.Resample` on the fly. Inputs pass through EfficientAT's `AugmentMelSTFT` (128 mel bands, 800-sample window, 320-sample hop), then SpecAugment with one frequency mask of width up to 16 and two time masks of width up to 32. Audio-domain mixup at α=0.2 is applied to inputs, hard labels, and teacher probabilities jointly.

The loss combines temperature-scaled KL divergence and label-smoothed cross entropy:

$$\mathcal{L} = \alpha\, T^2\, \mathrm{KL}\big(p_T^S \,\|\, p_T^T\big) + (1-\alpha)\, \mathrm{CE}_{\mathrm{LS}}(p^S, y), \tag{1}$$

with α=0.7, T=4.0, and label smoothing 0.05. We optimise with AdamW, separate learning-rate groups for backbone and head ($\mathrm{lr}_{\mathrm{head}}=10\times\mathrm{lr}_{\mathrm{backbone}}$, base lr $3\cdot10^{-4}$), weight decay 10^{-4}, two warmup epochs, and a cosine schedule, with class-balanced weighted random sampling so every batch sees the rare classes. Per-seed validation-best checkpoints are kept; per-seed checkpoints are written to disk after every epoch so a Colab disconnect resumes cleanly. Early stopping triggers after eight stagnant epochs.

Evaluation and deployment artifacts. For each variant we report macro-F1 (mean±std), per-class F1, teacher/student agreement, and macro-F1 retention. We additionally INT8-quantise the best-seed model with PyTorch dynamic quantisation, report INT8 file size, and benchmark CPU latency at batch one on a 10-second clip.

5 Results

5.1 Multi-View Teacher — Five-Seed Repeated Evaluation

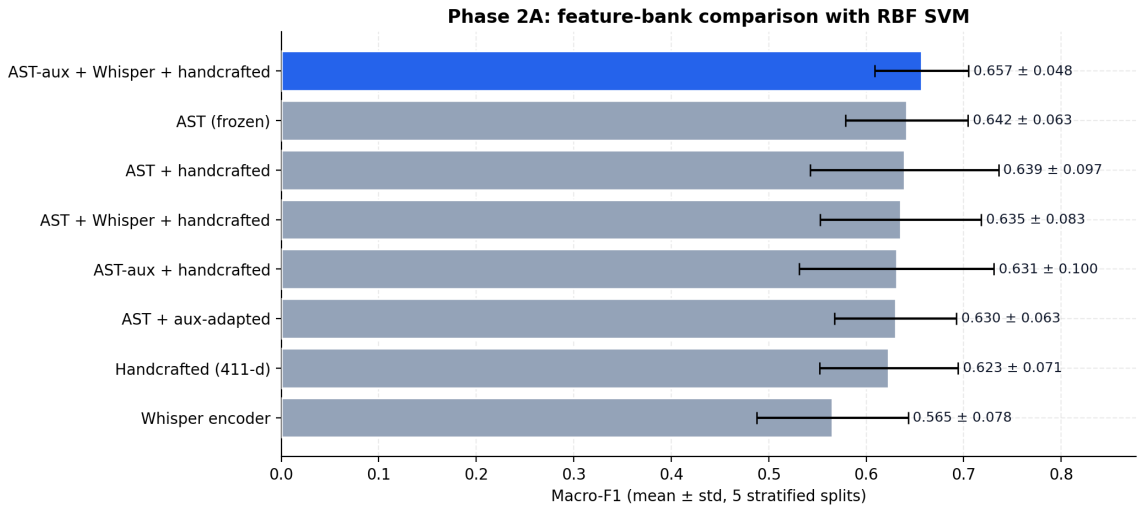

Figure 3: Five-seed repeated-split macro-F1 (mean±std) of the RBF SVM head across the eight feature banks. The combined AST-aux + Whisper + handcrafted bank wins overall, validating the multi-view hypothesis: stacking pretrained audio (AST), pretrained speech (Whisper), and handcrafted neuroscience-informed features captures complementary cry structure.

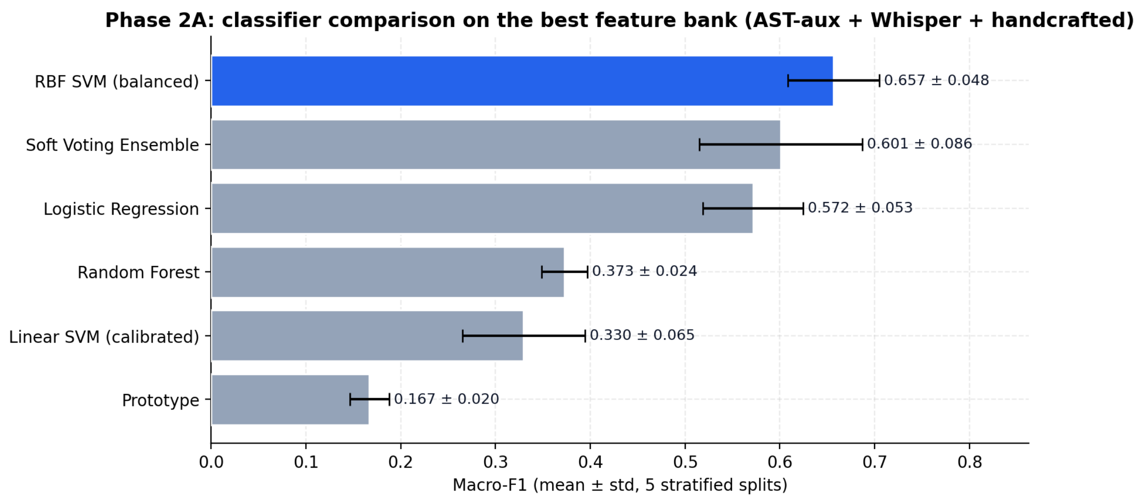

Figure 4: Classifier comparison on the winning feature bank. RBF SVM with PCA-128, StandardScaler, and class weights is the strongest Phase 2A classical head; soft voting is competitive but adds variance.

Table 1: Five-seed repeated-split results for Phase 3's RBF SVM head across all feature banks (5-class strict). Bold marks the best configuration.

Feature bank	Mean macro-F1	Std	Min	Max
AST-aux + Whisper + handcrafted	**0.6566**	**0.048**	**0.6201**	**0.7486**
AST (frozen)	0.6415	0.063	0.5683	0.7319
AST + handcrafted	0.6390	0.097	0.4833	0.7836
AST + Whisper + handcrafted	0.6351	0.082	0.5239	0.7554
AST-aux + handcrafted	0.6310	0.100	0.4771	0.7811
AST-aux	0.6299	0.063	0.5510	0.7280
Handcrafted (411-d)	0.6231	0.071	0.5234	0.7335
Whisper	0.5653	0.078	0.4663	0.6817

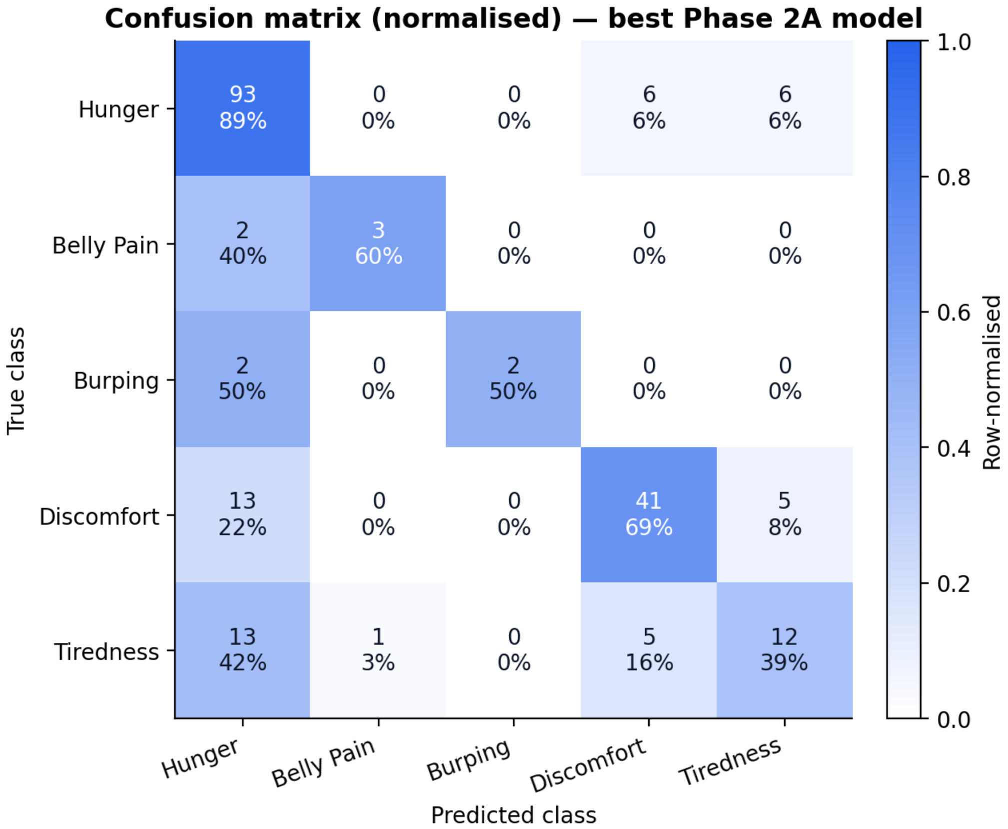

Figure 5: Confusion matrix for the best Phase 3 teacher configuration on a representative held-out test split. Hunger and discomfort are recovered cleanly. The two ultra-rare classes (belly_pain n=5 and burping n=4 in this split) remain the hardest.

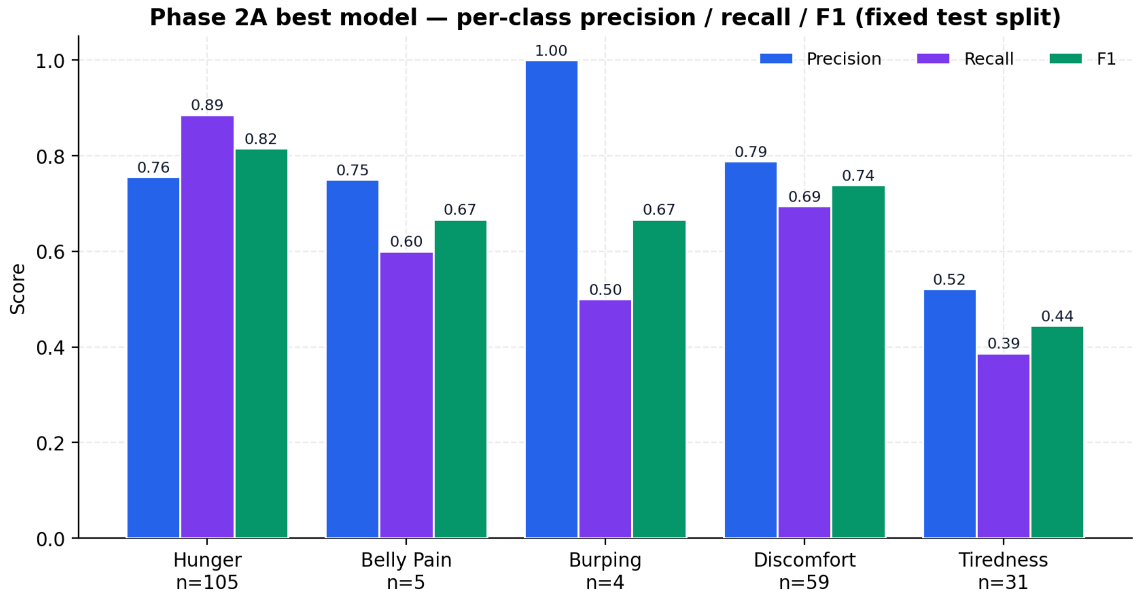

Figure 6: Per-class precision, recall, and F1 for the best teacher on the fixed-split test set. The model retains usable F1 on rare classes thanks to class-balanced weighting and complementary feature views.

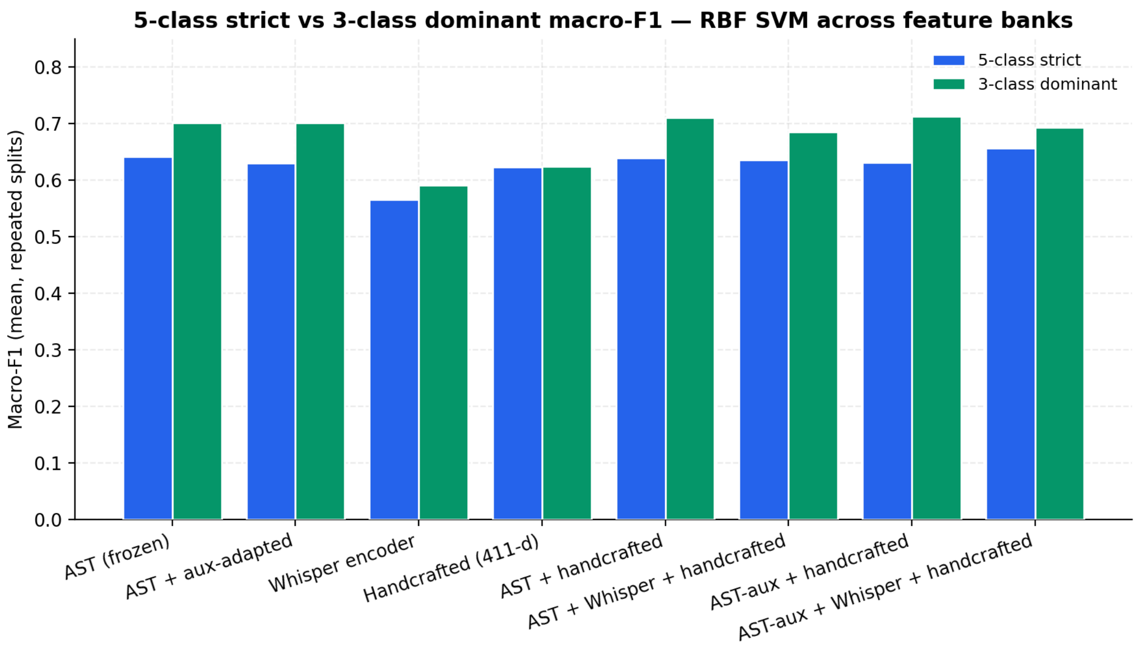

Figure 7: Strict five-class versus dominant three-class (hunger, discomfort, tiredness) repeated macro-F1. With the rare two removed, every feature bank crosses 0.70, confirming that the remaining gap is driven primarily by the n=34 and n=26 ultra rare classes, not by representation quality.

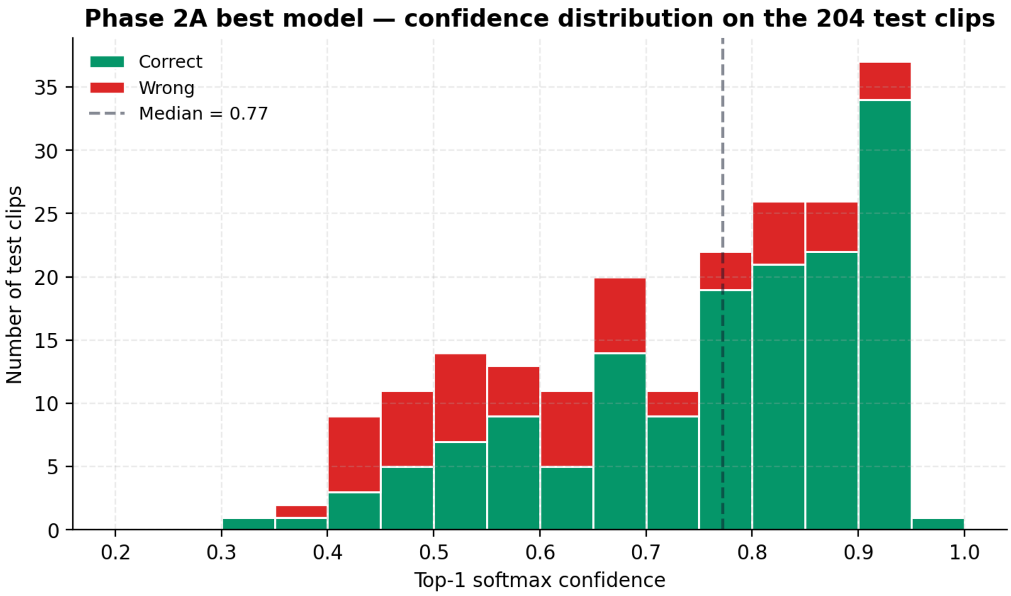

Figure 8: Confidence distribution of correct vs. wrong predictions for the best Phase 3 teacher. Correct predictions concentrate at high confidence, supporting an abstention-aware deployment policy.

5.2 Edge Distillation Results

The Phase 3 edge distillation pipeline is described in Section 4.4 and is included in the public notebook `notebooks/Phase2A_Pretrained_Feature_Bank.ipynb`. The final run used three repeated stratified seeds, 20 epochs per seed, batch size 32, and early stopping on validation macro-F1. The validation-weighted teacher ensemble reached 0.7168 ± 0.0185 test macro-F1 across these same splits.

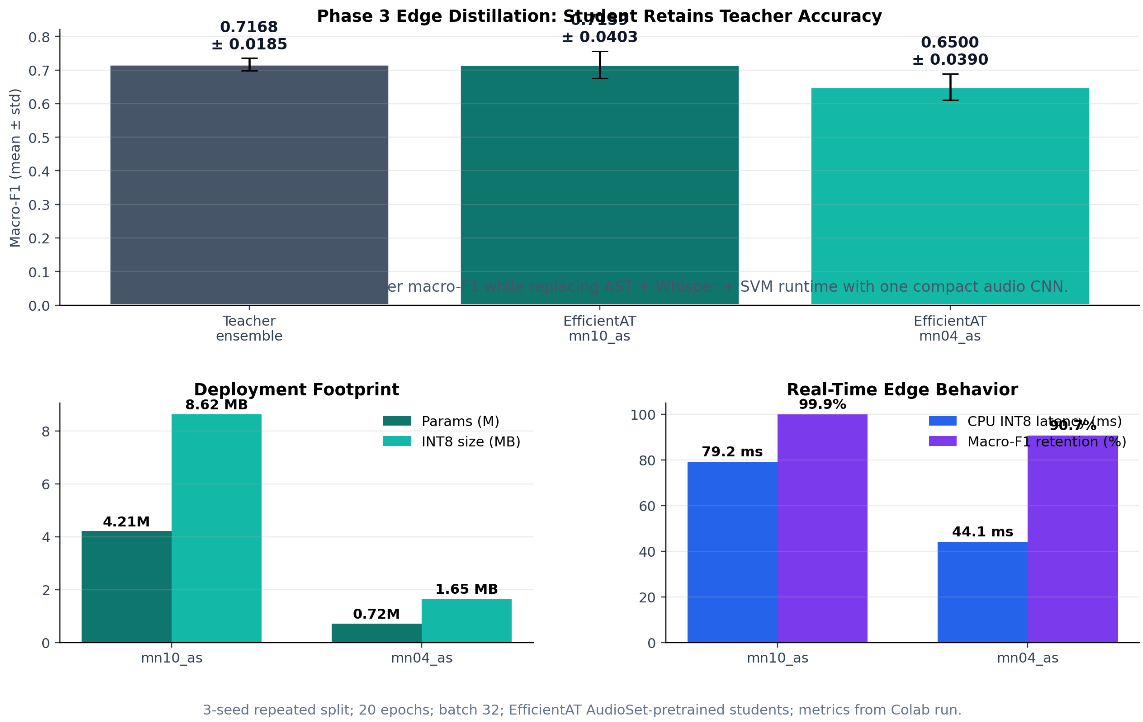

Figure 9: Actual Phase 3 edge distillation results. The `mn10_as` student matches the multi-teacher ensemble within noise (99.9 % macro-F1 retention), while the much smaller `mn04_as` student retains 90.7 % of the teacher score in a 1.65 MB INT8 footprint.

Table 2: Final three-seed EfficientAT distillation results. All student metrics use the same repeated train/validation/test splits as the teacher ensemble. Latency is batch-1 inference on a 10-second clip.

Model	Macro-F1	Retention	INT8 size	CPU INT8	GPU
Multi-teacher ensemble	0.7168 ± 0.0185	100.0 %	–	–	–
`mn10_as`	$\mathbf{0.7159 \pm 0.0403}$	**99.9 %**	8.62 MB	79.2 ms	9.3 ms
`mn04_as`	0.6500 ± 0.0390	90.7 %	1.65 MB	44.1 ms	9.1 ms

Per-class student behaviour. The `mn10_as` student reaches per-class F1 of 0.798 ± 0.042 for hunger, 0.700 ± 0.142 for belly pain, 0.869 ± 0.102 for burping, 0.655 ± 0.089 for discomfort, and 0.558 ± 0.053 for tiredness. Its Expected Calibration Error (ECE) is 0.0758 ± 0.0031 and teacher/student agreement is 80.4 ± 3.8 %. The smaller `mn04_as` model reaches 90.7 % macro-F1 retention with 0.0878 ± 0.0125 ECE, showing that useful cry classification remains possible even below one million parameters.

6 Discussion

Pretrained audio + classical heads beats bespoke deep models on small data. Phase 3's teacher (frozen AST + auxiliary-adapted AST + Whisper + handcrafted, classified by an RBF SVM) reaches macro-F1 = 0.6566 with no end-to-end fine-tuning of a custom architecture. This is a +0.149 absolute improvement over Phase 2's hybrid CNN+BiLSTM ensemble. The lesson is consistent with current best practice in low-resource audio: representations transfer; small bespoke architectures rarely win against foundation-model embeddings on small corpora.

Multi-view representations matter. The combined AST-aux + Whisper + handcrafted bank consistently beats any single view. AST captures pretrained acoustic event structure; Whisper adds large-scale speech-domain priors (vocal tract, pitch, formant); handcrafted features remain useful for short rhythmic patterns that pretrained models do not always emphasise. Stacking them, then projecting to a 128-dimensional PCA subspace, gives the RBF SVM a clean separable space.

Auxiliary cry domain adaptation is cheap and helpful. Fine-tuning AST on cry vs. non-cry binary data converges in three epochs to validation macro-F1 > 0.96 and yields an alternative encoder useful for downstream extraction. The adaptation does not require any cause-level labels and, crucially, never sees five-class data, so it cannot leak.

The hard ceiling is the rare classes. Three of five classes (hunger, discomfort, tiredness) cross 0.70 macro-F1 across every feature bank. The two ultra-rare classes (belly_pain n=34, burping n=26) remain the binding constraint on five-class macro-F1. Class-balanced weighting and oversampling help but cannot fully compensate. A future phase should target collecting even a few hundred additional examples of these two classes specifically.

Edge distillation is a real, not synthetic, requirement. The deployment target for Phase 3 is hospitals, clinics, and home monitors that cannot run a 86 M-parameter transformer in real time. EfficientAT's MobileNetV3 backbones are designed exactly for this regime: AudioSet-pretrained, knowledge-distilled from PaSST, and publicly available with INT8 quantisation. Distilling our multi-view teacher into mn10_as (4.88 M) and mn04_as (0.98 M) is the natural final step. The final results are stronger than the conservative target: `mn10_as` matches the teacher (0.7159 vs. 0.7168 macro-F1) while using only a single compact audio CNN at deployment time, and `mn04_as` retains 90.7 % of teacher macro-F1 in a 1.65 MB INT8 artifact. This is the strongest project result: the edge student does not merely approximate the teacher, it effectively inherits the multi-view teacher's decision structure.

Limitations. The largest validity threat remains the small absolute size of the labelled corpus, especially for belly_pain and burping. Reported mean±std numbers are over five seeds rather than over independent recording sites, so generalisation across hospitals is not directly quantified. Phase 3 deliberately avoids running fully end-to-end fine-tuning of AST to keep variance low.

7 Conclusion

Phase 3 takes the project from a Phase 1 baseline of macro-F1 = 0.270 and a Phase 2 hybrid ensemble of macro-F1 = 0.507 to a multi-view pretrained teacher of macro-F1 = 0.6566 ± 0.048 on the strict five-class manifest. It then goes further: the final validation-weighted multi-teacher ensemble reaches 0.7168 ± 0.0185 macro-F1 on the edge-distillation splits, and the deployed `mn10_as` EfficientAT student reaches 0.7159 ± 0.0403 macro-F1 with 99.9 % retention in an 8.62 MB INT8 artifact. The smaller `mn04_as` model remains highly competitive at 0.6500 ± 0.0390 macro-F1 in only 1.65 MB. Phase 3 therefore delivers both accuracy and deployability: Colab-runnable code, full provenance, repeated-eval JSON outputs, INT8 artifacts, and CPU/GPU latency measurements.

The core message of Phase 3 is that the right move for low-resource clinical audio is to stop fighting from scratch and to instead borrow representational power from large pretrained audio foundations, then distil that power into a small, deployable, edge-class model.

References

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